POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2015
Programme: BE | Full Marks: 100
Course: Object Oriented Software Engineering | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Define Software Engineering and its significant role over the system design. 7
b) Define agile software development with its advantages and disadvantages? 8
c) Define risk. Explain how risk management is carried out in a software project? Explain all the steps. 7
d) Explain why software system which is used in a real world environment must change or become progressively less useful. 8
e) Obtain 1-level DFD for Movie management system: 8
A customer can book a ticket from the internet or can directly buy the ticket in the Movie Hall itself. There can be multiple halls within one movie theatre. The ticket operator provides a ticket with hall's stamp after checking the booking information to the customer. The guard in each hall validates the ticket and provides access to the customer inside the hall. There is also provision of complementary food item which the café will provide in the break time of the movie.

b) From the same above mentioned case, obtain a use case diagram also. 7
a) What is integration testing? Differentiate between top- down and bottom- up integration testing. 8
b) List out the differences between quality control and quality assurance and explain the elements and goals of software quality assurance (SQA). 7
a) What is SPI and SPI framework? Explain briefly about CMMI. 8
b) Define requirement engineering. What are various methods of requirement elicitation? Explain. 7

6. a) What is difference between class diagram and object diagram? Draw class diagram for the following scenario. 8
A bank provides debit card service for its customers only for current account and saving account. Customers can perform withdraw, transfer, balance query and pin change operations.Customers can use ATM of any location.
b) What are test cases and what is the importance of testing and designing test cases. 7

7. Write short notes on: (Any two) 2×5
a) Six Sigma
b) Functional Independence
c) Deployment level design.

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